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ELEN4006 – Measurement Systems

The design, building, and testing of a displacement measurement system to monitor and characterize hand motion capability for stroke rehabilitation.

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Abstract:

Stroke rehabilitation is a program of different therapies which aims to reduce disability and promote participation in activities of daily living in individuals who have suffered from stroke. This report discusses the design, construction, and testing of a displacement measurement system to monitor hand motion capability. The system is tested within a controlled clinical environment that will assist clinicians to characterize the rehabilitation progress of stroke patients with regards to using their hands. However, the approach to design of the system is confined to just monitoring and characterising if whether movement has occurred adequately based on a 20mm threshold due to the limitation of the sensor. The SS495A Hall effect sensor is used as the sensing element which interprets displacement relative to the change of the magnetic field. Signal conditioning is unnecessary in this design as the sensor output voltage range of 2.5V - 4.82V, is within the input range for the microcontroller in the signal processing stage. An Arduino Uno microcontroller is used for signal processing achieving analogue-to-digital conversion of the signal for display on a liquid crystal screen, whilst also correcting the non-linearity of sensor. Overall, the system has an error of 14.17%. and operates at a bandwidth of 181.3Hz. Further recommendations to use a more robust sensor are suggested in order to improve the system to accurately characterize the various hand moves above just knowing that movement has just occurred.

1. INTRODUCTION

Effective stroke rehabilitation relies on accurate monitoring of motor recovery progress, particularly in hand function, which is critical for patients to regain [1]. To address this requirement, this report presents the design, construction, and testing of a displacement measurement system that detects hand motion using a Hall effect sensor. The system relies on detecting changes in the magnetic field from a magnet placed on the patient's hand to indicate the movement of the hand. The design does not require signal conditioning as the sensor voltage range is withing the 5V limit for signal processing, an Arduino Uno microcontroller is used for signal processing. The result of signal processing is shown on a I2C display module which shows the displacement as well as corresponding classification of hand movement. The designed system characterizes horizontal movement of the hand based on wrist extension [2]. The report is structured as follows, section 2 presents the design specifications and high-level solution, with the detailed design in section 3. The performance evaluation is presented in section 4, followed by critical analysis in section 5. The reflection of group work and summary and conclusions are presented in sections 6 and 7 respectively.

2. DESIGN SPECIFICATIONS AND HIGH-LEVEL SOLUTION.

The specifications of the designed measurement system are shown in Table 1. The system is used in the context of classifying adequate and insufficient movement based on how much the hand has moved from an initial position. Figure 1 shows how the system will be used in the real- world. The high-level solution is shown in Figure 2.

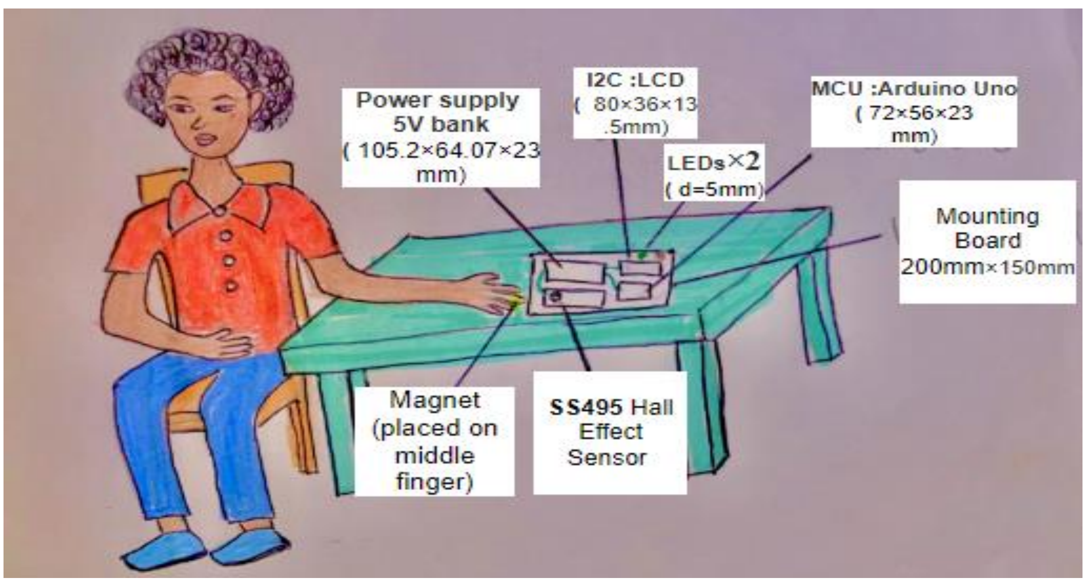

Figure 1: Diagram showing how displacement measurement system is used to detect hand movements.

Table 1: Displacement measurement system design specifications

Specification	Component	Specifications	Justification
Power supply (V)	Power bank	• 5V power supply from using Duracell Plus Alkaline Battery	Arduino has an input range of 0-5V range.[3] The HES saturates within these ranges for 5V supply. Ensures optimal sensor output range (2.55V–4.85V per testing)
Input range(mm) 10mm-40mm	Hand	• Input range (true variable): ○ 10mm-40mm in displacement	Effective sensing region of the sensor relative to the hand starting at 10mm.
Output range(V) 2.5V -4.84V	LCD	• Adequate movement (corresponds to a displacement in range 20- 30mm) which relates to voltages of • Inadequate movement (corresponds to a displacement in range 10-19mm)	Adequate movement is a wrist flexion at normal degree deviation movement is at 90º[4] which corresponds to a displacement within a range of 20- 30mm , whilst extension is limited to 40 mm corresponding to 3.94V
Display	LCD Screen	• Using 3 grids for acronym characterization • fresh rate of 60 Hz	Displaying the characterization of the movement at an acronym based on a displacement of two levels. ADQ (adequate) INS(inadequate)
Temperature	Testing Room	• 20º -40º room temperature	Expected room temperature levels in Johannesburg[5]
Linearity	SS495	• Ideal linearity of the sensor: y=0.075x+1.77	Fitted line of "best-fit" which is ideal linearity based on the recorded measurements and calculated to determine linearity[6]
Error (%)	Sensor (3%) MCU (2%)	• Difference from linearity at 3% error • MCU sampling and processing at 2% error	Total acceptable error at 5% which will ensure accuracy in displacement [7]
Steady-state sensitivity **K**	Ideal linearity of sensor data	• Sensitivity = 0.075V/mm	This gives the effective range that our system is able to effectively interpret the displacement input within
Bandwidth (Hz)	97Hz	• Sampling rate of MCU 10 samples than averaging • Computational time for outputting to LCD	Since hand moves relatively slowly it is not the benchmark for the bandwidth but rather the MCU sampling rate as well as computation speed[8]
Hysteresis (% of f.s.d)	2%	• There is a slight difference relative to the input changes. The sensor had 2%	Therefore negligible, since only expect linear movement in one dimension

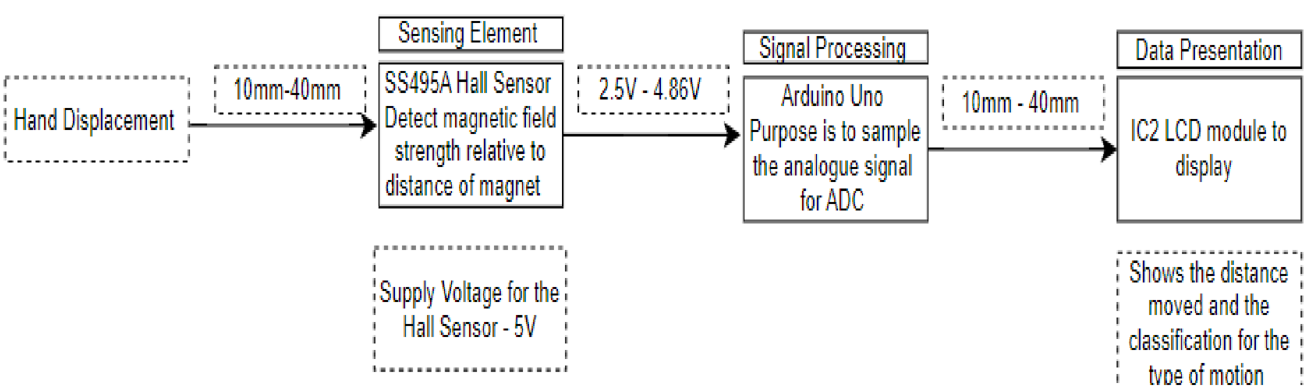

Figure 2: Bentley's model High-level solution for Displacement measurement system for hand motion

3. DETAILED DESIGN

Table 2 presents the justification of the design choices and component choices used in figure 2 showing the integrated system circuit diagram.

Table 2 : Detailed Design Justification

System aspect	Component Choice	Design Specifications & calculations	Justification
Bandwidth	-	• Since the bandwidth of the sensor is at 86Hz estimate a slight increase based on Arduino computation and sampling • BW 90Hz	Calculated based on consideration that the processing might increase the bandwidth if the system slightly from that of the sensor [9]
Cost	-	• R300	This is a justified cost based on simple motion characterization
Accuracy		• Accuracy - 95%	Since working with such small displacement
Sensor	Hall effect sensor	• Input range (true variable):10mm-30mm in displacement	Effective sensing region of the sensor relative to the hand of average length 20cm observed at 10mm away
Signal processing	Arduino Uno Resolution 1023(10 bits)	• Non-linearity correction: subtracting 2.45Vconverts this voltage to a digital value (0–1023 for a 10-bit ADC) for calibration and analysis: y=0.075x+1.77 • Interpretation of voltage level as actual displacement measured at threshold =20mm: Insufficient (INS) adequate (ADQ)	The sensing element is non-linear but assume linear relationship for processing [6] Correct for the displacement reference of 10mm of the hand is placement relative to system [10] Compute the displacement based of the voltage produced relative to the field, either by using look-up table or linear equation [11]
Data presentation	LCD Screen Green LED Red LED	• Using 3 grids for acronym characterization Insufficient (INS) displacement <20mm Adequate (ADQ) displacement ≥20mm • fresh rate of 60Hz	Displaying the characterization of the movement at an acronym based on a displacement of three levels.

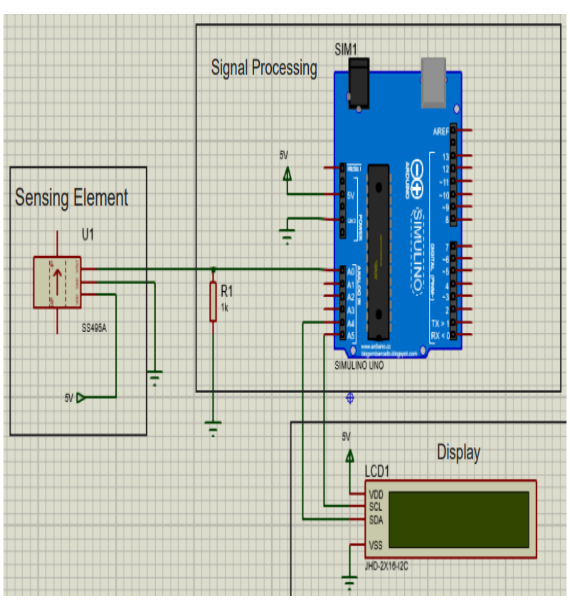

Figure 3: Design system circuit diagram.

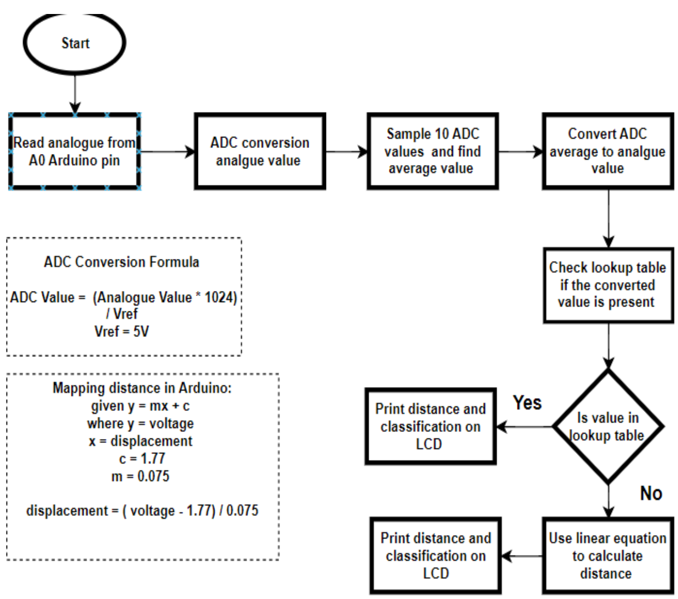

Figure 4: Microcontroller's logic flow chart.

4. PERFORMANCE EVALUATION

The designed system performance evaluation is carried out in this section. The measurement methods used to obtain results for each stage are explained in Table 3. The methods explained were used to obtain the results shown in Table 4. Figure 5 shows the system step response.

4.1 Determining Overall system error:

Table 3: System measured variables and computed error per stage analysis

Input	Measured Values at each stage			Calculated error			
Displacement (mm)	**Sensor (V)**	**Signal Processing (V)**	**Display (mm)**	**Sensor (%)**	**Signal Processing (%)**	**Display (%)**	**Total System (%)**
10	4.85	4.82	9.89	0.78	0.83	0.29	1.9
13	4.64	4. 62	12.19	1.14	0.76	0.13	2.03
16	3.37	2.96	17.32	0.67	0.96	0.34	1.97
19	2.81	2.68	19.45	0.85	0.48	0.28	1.75
22	2.74	2.69	21.97	1.74	0.63	0.15	2.52
25	2.62	2.53	24.16	0.72	0.32	0.67	3.53
28	2.50	2.46	27.92	0.58	0.32	0.41	1.27
31	2.52	2.48	31.89	0 .32	0.84	0.66	1.82
34	2.52	2.48	33.98	0.61	0.47	0.72	1.80
37	2.52	2.48	36.76	0.34	0.63	0.46	1.43
40	2.52	2.48	39.64	0.82	0.78	0.23	1.83

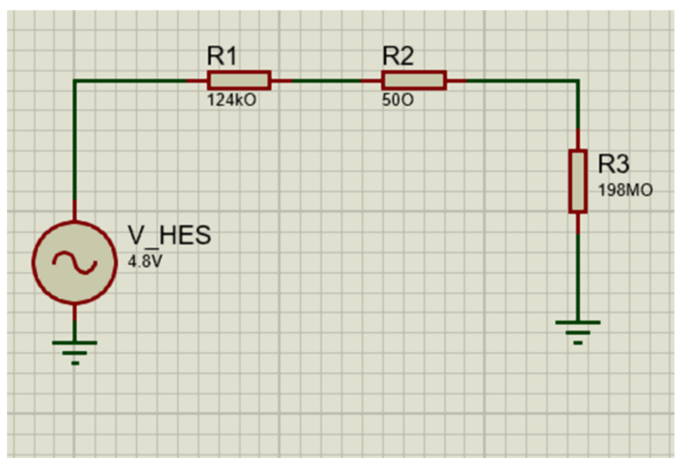

Figure 5: The system input-time series response. Figure 6: Equivalent circuit for Loading Effects

4.2 Analysis of loading effects:

Loading effects occur when the measurement system itself alters the signal it is measuring, typically due to impedance mismatches or power draw [12]. Figure 6 illustrates the equivalent circuit for loading effects analysis, the experiments result allowed to quantify the impedance in the sensor as 124kΩ and the Arduino impedance as 198MΩ. R_2 represents the cable impedance which is assumed as 50Ω

5. CRITICAL ANALYSIS

5.1 Suitability of system

The system offers several advantages, including low cost, non-invasive operation, and real-time feedback, making it suitable for stroke rehabilitation monitoring. However, it is limited to one axis of motion and requires individual calibration for each patient. Performance testing revealed a total system error of 14.17% calculated using the RMS method[13], which exceeds the target error of 5%. Despite this, the system demonstrated reliable sensitivity within the 10–40mm displacement range, a bandwidth of 181.35Hz which is higher than the design specifications, and an accuracy of 85.83% which is lower than required when compared to table 2.

Table 4: Total system error based on each stage.

Stage	**Sensor**	**Signal processing**	**Display**	**Total**
Error (%)	9	3	2.17	14.17

5.2 Cost analysis

The system's overall cost of R388.40 is shown by the cost breakdown in table 5. The overall cost deviates from the estimated cost in table 2 by R80.0. However, it can be concluded that the designed system is still affordable.

Table 5: The Designed system cost analysis

Components	**HES**	Magnets	**Arduino Uno**	**LCD**	**Batteries**	**Total cost**
Price	R125	R18.40	R105	R60	R80	R388.4

5.3. *Future recommendations*

For improved measurements for better hand motion characterization, it would be to place more magnets on the hand to enhance field strength. Characterize better thresholds based on more functional wrist actions, like extension and flexion using a more sensitive sensor.

6. REFLECTION OF GROUP WORK

A detailed time management plan which consists of weekly deliverables was created and followed. Marcus designed, built, and tested signal conditioning. Sipho designed, built, and tested signal processing. The overall system performance results were collected as a group and report writing.

7. CONCLUSION

The system measures finger displacement (10–40mm) at 85.83% accuracy and 181.35Hz bandwidth. While its 14.17% error exceeds the 95% accuracy target, due to non-linearity correction in the signal processing, it remains a cost-effective solution for stroke rehabilitation monitoring.

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